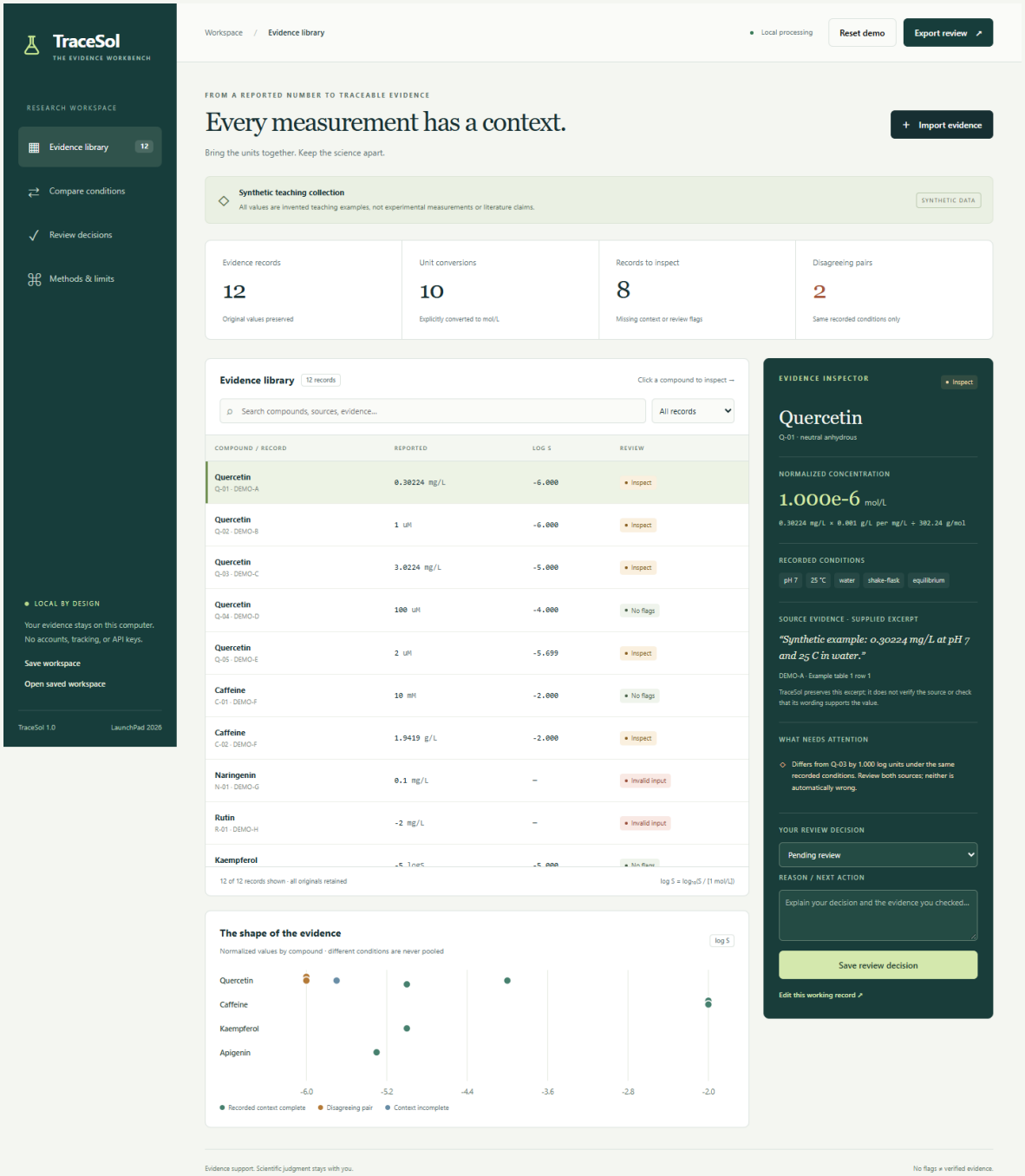
Every measurement has a context.

TraceSol helps chemistry students and researchers normalize solubility units, inspect source evidence, and separate disagreements from differences in experimental context.

Local processing / No API keys / No pip dependencies / No video required



Actual application screenshot. All displayed measurements and excerpts are synthetic teaching examples, not published experimental results.

A review you can reproduce.

Import a CSV, preserve the original record, inspect each conversion and source excerpt, then add your own decision and export the evidence trail. The application never chooses a source as correct or declares a record model-ready.

The useful three-minute demonstration

1. Equivalent units. Q-01: 0.30224 mg/L / 302.24 g/mol = 1 micromolar. Q-02 reports 1 micromolar directly. Both normalize to log S = -6.

2. Context-aware inspection. Q-03 is tenfold higher under the same recorded conditions; two pairs are flagged. Q-04 has different pH and Q-06 a different form, so these stay separate. Missing pH prevents a complete-context conclusion.

3. Human reasoning. Add a follow-up decision and a reason. Edit a working value to see the comparisons recalculate and its decision reset. Export the review or save/reopen the workspace.

Implemented and checked

Area	Evidence in the package
Calculation & server	32 automated tests; standard-library unittest suite
Browser workflow	15 checks using Playwright with installed Microsoft Edge
Responsive layout	Checked at 390, 768, 1024, and 1440 pixels
Deliverables	CSV, audit JSON, printable HTML, editable workspace
Source trail	Original fields, excerpts, locators, calculations, SHA-256

Transparent boundaries

The demo is synthetic. Source excerpts are not independently verified. Compound identities are user-provided labels. Matching recorded conditions does not establish full experimental equivalence. No paper-validation accuracy, predictive performance, or time-saving claim is made.

Run: extract the release; double-click START_TRACESOL.cmd on Windows, or run python app.py with Python 3.10+. Open http://127.0.0.1:8765. No internet is required after Python is available.

Submission state: local build and written materials prepared; not published or submitted. Confirm age/student/geographic eligibility and add a public downloadable-build URL accessible through judging. Video is optional under the public event requirements.

Background: IUPAC Gold Book entries S05740 and A00295; Llompart et al., Scientific Data (2024), DOI 10.1038/s41597-024-03105-6. Full methods, credits, and validation are included in the project.